KO LIH HAN

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EDUCATION

National Tsing Hua University Jun 2026
M.S., Information Systems & Applications | GPA 4.09/4.3

Chung Hua University Jun 2024
B.S., Computer Science & Information Engineering | GPA 3.9/4.0

RESEARCH EXPERIENCE

Interpreting Implicit Intent through Workplace Hierarchies and Context M.S. Thesis
National Tsing Hua University | NLP, retrieval, Role/Context KGs, Llama 3.1-8B

- Curated 20,455 workplace utterances from 7 forums with speaker/listener state, emotion, and purpose labels to study context-dependent implicit intent.
- Evaluated Role KG + Context KG retrieval with Llama 3.1-8B on 400 targets, improving semantic F1 from 0.644 to 0.682.

SELECTED AI PROJECTS

Small-Agent QLoRA

Python, Qwen3.5-9B, QLoRA / PEFT, Transformers, llama.cpp

- Built a tool-using agent with search, file inspection, Python execution, timeouts, duplicate blocking, and trace logs.
- QLoRA on corrective failure examples raised 200-task success from 34.5% to 62.0%.
- Evaluated on GAIA100, raising accuracy from 9% to 18%; failure analysis showed over-search remained the main failure mode.

Hybrid Retrieval Evaluation

Python, BM25, Qwen embeddings, RRF, FastAPI

- Built a hybrid retriever combining BM25, Qwen dense retrieval, RRF, and inspectable rank provenance.
- Evaluated 64 queries over 5,189 documents; Recall@10 rose from 0.734 to 0.763 and MRR@20 from 0.679 to 0.742.

Semantic Text-to-SQL

Python, FastAPI, LangGraph, SQLite, Qwen3.5-9B, pytest

- Built an NL-to-SQL service with schema grounding, validation, bounded repair, and read-only execution.
- Shared one service layer across FastAPI, CLI, Streamlit, and evaluation to prevent path drift.
- Evaluated 100 BIRD DEV queries; correct-query accuracy rose from 33% to 39%, fixing 6 queries with no regressions.

SOFTWARE ENGINEERING EXPERIENCE

Software Developer Aug 2022 – Aug 2026
Kahap Inc | Part-time

- Built and launched a feedback platform for submissions, discussions, admin review, and notifications.
- Maintained a Laravel/Vue.js production platform across queues, scheduled jobs, migrations, and Linux operations.
- Integrated GPT-SoVITS, pronunciation checks, and text checks into backend speech workflows.

Full-Stack Developer Intern Jun 2021 – Jan 2022
Pac Com Solution

- Built features and fixed bugs in a production Laravel/Vue.js POS system.

SKILLS

AI / LLM:	Agents & tool use, RAG / hybrid retrieval, evaluation, PEFT / QLoRA, Transformers, LangGraph, llama.cpp
Backend:	Python, FastAPI, PHP, Laravel, REST APIs, SQL, MySQL, SQLite
Web & Tools:	JavaScript, Vue.js, Linux, Git, pytest
Languages:	Chinese (Native), English (Fluent), Malay (Intermediate)